Planting a Solution and Watching the Learner Miss It: the 2026-09 planted-teacher campaign

the snn-research program

2026-09-08, v2.1 (Phase 2b addendum gated)

Abstract

Every lab-scale "capacity" number this program has produced for the tempotron was a *findability* wall: the load $\alpha = P/N$ at which Adam-cosine gradient descent, run for a fixed budget, stops finding a solution on half the random-label problems. Whether a solution still existed above that wall was never certified at $N = 500$. This campaign labels the patterns with a random-weight *teacher* tempotron, so that a solution exists at every load by construction, and asks two pre-registered questions on $n = 32$ identified tasks per rung: **Q1** does the wall move when a solution is planted, and **Q3** does the comb kernel's lift over the sinc kernel survive planting. In both count-fixed ensembles (one spike per afferent; exactly two) the planted low wall sits within one resolution atom of the random wall ($\Delta_{\text{low}} = +0.029\,[-0.015, +0.097]$ and $+0.008\,[-0.038, +0.065]$), consistent with a learner wall rather than a solution-space wall (H-learner). The planted curve then *re-solves* above a hard band ($29/32, 30/32, 32/32, 32/32$ at $\alpha = 6, 8, 12, 16$ against $0/32$ for random labels), and the students that re-solve have recovered the teacher (overlap $q \geq 0.80$ on every re-solved task, 0.999 at $\alpha = 16$), while the students that solve at or below the wall are unrelated to it ($q \approx 0$, decision times at chance). The comb kernel's planted problem never walls (≥ 16), so the comb–sinc split survives planting with a lower bound $D_3 \geq +12.2$; the random lift itself is $+0.35$ and $+0.34$ in the two count-fixed ensembles and $+0.50$ under Poisson counts. A perturbation ladder around the teacher shows that gradient descent leaves the planted solution immediately and re-solves nearby (overlap ≈ 0.8, norm 3.5–8 $\|w_T\|$) from every start within $r = 2$ of it, at and above the wall, where random starts never solve. Norm-matched replicates on both cells show that the collapse of the un-matched ladder below the random-init rate at $r = 8$ was the inflated starting norm; with the norm matched, a start at overlap 0.13 solves at a rate not distinguishable from random init, and the edge moves out by about a factor two (partly a norm effect). Four hypotheses died on registered kills, including the Poisson-count "DC mode" as the explanation of the comb lift. Spend $\approx$ \$136 of \$170 (census through Phase 2b); every number was recomputed by an independent verifier before it was written down.

Contents

1 Background and theory

1.1 The object

A tempotron [1] is a leaky integrate-and-fire readout of N afferents with weights $w \in \mathbb{R}^N$. Afferent i emits spikes at times $\{u_{i,f}\}$ in a window $[0, T]$; each spike contributes a kernel $K(t - u_{i,f})$ to the trace $x_i(t)$, and the membrane potential is $V(t) = \sum_i w_i x_i(t)$. The neuron fires on a pattern if $\max_t V(t) \geq \theta$. A pattern set of P patterns with binary labels $\sigma \in \{\pm 1\}^P$ is *solved* if the neuron fires on exactly the positive ones. The load is $\alpha = P/N$. Three kernels are used at matched effective temporal dimension $K_{\text{eff}} = T/\tau \approx 32$ on a 64- or 75-bin grid: the double-exponential of Gütig–Sompolinsky (dexp, $\tau_s = \tau_m/4$), a sinc (band-limited, non-negative integral), and an odd-harmonic comb ($m = 4$ Lanczos-windowed cosines, $\int K = 0$). The comb is the kernel whose findable capacity had shown an unexplained +0.45 lift over sinc in 2026-08 [5].

1.2 Existence versus findability

The version space of a labelled pattern set is the set of w that solve it. Because the readout is a maximum over time, the version space is a union of convex cells indexed by the assignment of a decision time to each positive pattern; above a critical load it shatters into exponentially many small cells [2]. The program's lab measurements were all *findability at a budget*: the first downward half-crossing of the solve fraction of an Adam optimiser with a cosine learning-rate schedule (learning rate 0.1, batch 16, 8000 epochs, two restarts; the "E1 recipe") as α increases. No result at $N = 500$ carried an existence certificate; the toy scale ($N = 16$) had shown random labels provably infeasible where the learner had long stopped solving, but also that the learner stops before existence ends.

1.3 Planting

Draw a teacher $w_T \sim \mathcal{N}(0, \sigma_w^2 I)$, compute its $V_{\max}$ on every pattern, set the teacher threshold θ_T at the median of $V_{\max}$ (the split rule for odd P), and label $\sigma_\mu = \text{sign}(V_{\max,\mu} - \theta_T)$. The labels are then exactly balanced, and w_T solves the problem at every α. This is the planted-ensemble construction of constraint-satisfaction theory [3]. Two facts from the theory pass (THEORY.md, T1 and T5) frame what to expect:

- **Size biasing (T1.1, exact).** With Ω the weight space modulo scale and $V(\sigma)$ the volume fraction of the version space of σ, a planted label vector is drawn with probability $V(\sigma)$, so for any $f \geq 0$,
$$\mathbb{E}_{\text{plant}}[f] = \frac{\mathbb{E}_{\text{rand}}[fV]}{\mathbb{E}_{\text{rand}}[V]}, \qquad \frac{\mathbb{E}_{\text{plant}}[V]}{\mathbb{E}_{\text{rand}}[V]} = 1 + \mathrm{CV}^2_{\text{rand}}(V).$$
The planted version space is the size-biased, atypically fat cell, not the typical one. Its sign predicts a planting bonus $b > 0$ on *existence*; it says nothing about what a gradient learner finds.

- **Two loads (T5.1).** α_{onset}, where the student's overlap with the teacher $q = \cos(w, w_T)$ first leaves its background, obeys $\hat{\alpha}_f(\text{random}) \leq \alpha_{\text{onset}} \leq \alpha_c(\text{random})$: as long as the learner still solves random labels, far-from-teacher solutions exist and are findable, so the planted arm has no reason to be near the teacher; once no uncorrelated solution exists, q must leave zero. Full alignment ($q \to 1$) cannot occur below the random existence capacity. For the binary perceptron the analogous numbers are 0.83 (existence) against 1.249 (recovery) [4].

1.4 The replica-symmetric surrogate and the count law

The program's existence estimates came from a Gaussian surrogate: average over random w, so $V(t)$ is a Gaussian process with covariance $\propto \sum_i \mathbb{E}[x_i(t)x_i(s)]$. With k spikes per afferent at i.i.d. times,

$$\mathbb{E}[x(t)x(s)] = \frac{\mathbb{E}[k]}{T} D(t,s) + \frac{\mathbb{E}[k(k-1)]}{T^2} m(t)m(s), \qquad D(t,s) = \int_0^T K(t-u)K(s-u)\,du, \quad m(t) = \int_0^T K(t-u)\,du,$$

so, up to scale, $\mathrm{Cov}_V \propto D + g\,mm^\top/T$ with $g = \mathbb{E}[k(k-1)]/\mathbb{E}[k]$ (T4.1, exact). Spike-count *variability* injects a rank-one "DC" mode of relative weight $g\,c$, $c = (\int K)^2/(T\int K^2)$, with $c_{\text{dexp}} > c_{\text{sinc}} > c_{\text{comb}} = 0$: $g = 2$ for Poisson mean two, $g = 1$ for exactly two spikes, $g = 0$ for one spike. Two registered predictions followed: the smooth-kernel wall should depend on the count law (T4.3), and, as the campaign's first hypothesis H-DC, the comb lift over sinc should reverse in the one-spike ensemble if it were a DC-mode artefact (T3.2).

2 Design and pre-registration

2.1 Questions

Q1 (primary). Is the learner's wall insensitive to solution-space geometry? Statistic: $\Delta_{\text{low}} = \hat{\alpha}_{\text{low}}(\text{planted}) - \hat{\alpha}(\text{random})$, paired on identical spike draws, five bands at thresholds $\{-0.30, -0.10, +0.10, +0.30\}$ (PLANTED-HARDER, NOT RESOLVED, LEARNER-BOUND, NOT RESOLVED, GEOMETRY-SENSITIVE). **Q3 (primary).** Is the comb lift a learner or an existence property? $D_3 = G(\text{planted}) - G(\text{random})$ with $G = \hat{\alpha}(\text{comb}) - \hat{\alpha}(\text{sinc})$; a bound form $D_{3,\text{lb}}$ was registered before the comb cells were bought, for the case that the planted comb curve never falls. **Q2 (rider).** Routing or settling: does the learner fail because it cannot route to the teacher's decision-time assignment, or because it settles elsewhere? **Q4 (rider).** Does the student find the teacher or another solution: $q(\alpha)$.

2.2 Held fixed and moved

Held: $N = 500$, $T = 500\,\text{ms}$, $K_{\text{eff}} = 32 \pm 2\%$ per kernel and ensemble, the E1 recipe, threshold gauge (median $V_{\max}$; teacher split rule, shown KS-equivalent to the random gauge on 6/6 cells), label balance $f = \frac{1}{2}$, no margin. Moved, within level only: label mode (random, planted); kernel (comb $m = 4$, sinc; dexp dropped after Phase 0 for a learner-validity flag); spike ensemble (Poisson mean 2; exactly one spike per afferent; exactly two, "fixedk-2", which is density-matched to Poisson-2 but has $g = 1$). Every cell uses $n = 32$ identified tasks (seeds 0..31 of a deterministic stream; identity by pattern hash), a resolution atom of 0.10 in α, task-identity bootstraps with 2000 draws, and a resolved-transition requirement (a rung ≥ 0.9 below and ≤ 0.1 above the crossing).

2.3 Phases and gates

Phase 0 built the ports (teacher mode, one-spike and fixed-count ensembles, dexp), reproduced the archived anchors (comb 3.740 vs 3.747; sinc 3.250 vs 3.243; the mainline dexp recipe 3.461), calibrated the planting bonus at the toy scale where existence is exact, and measured the learner-validity gates the August campaigns had only listed: learning-rate robustness per cell (V4), budget stability (V5), gauge equivalence (V3). Phase 1 bought the $n = 32$ grid; Phase 2 the riders on the saved weights plus a perturbation ladder around the teacher; Phase 2b a norm-matched replicate of the ladder. Each phase had an independent pre-spend gate and an independent post-hoc gate (`numerics-verifier`); every landed number was recomputed from the raw rows before it entered the claim ledger, and a meta-review read across the gates at the Phase 1/2 boundary. Four hypotheses and one rider statistic were killed or struck on registered kills, and two instruments were found degenerate (§4).

3 Results

3.1 The walls: planting does not move the low wall (Q1)

Table 1 lists the gated half-crossings. The two paired contrasts return LEARNER-BOUND: the planted low wall coincides with the random wall to within one atom on both count-fixed ensembles, with the whole confidence interval inside the band. A solution exists at every load by construction, so the low wall is consistent with a findability wall of the learner (H-learner); the gate refused the stronger "learner phenomenon, not geometry". Under Poisson counts the planted sinc problem has no wall to $\alpha = 16$ and the contrast is not scorable; that ensemble's planted ease is booked on the lower-bound branch.

Table 1: Findable half-crossings at $n = 32$ (E1 recipe, 8000 epochs, harness `2632a91`), task-identity bootstrap 95 % intervals. "$\geq T$" is a grid bracket: solved 32/32 at every bought rung up to T; the rung above was not bought (costed gap).

kernel	ensemble	random $\hat{\alpha}$	planted $\hat{\alpha}_{\text{low}}$	Δ_{low} (paired)
sinc	one spike	3.411 [3.389, 3.436]	3.440 [3.412, 3.487]	+0.029 [−0.015, +0.097] **LEARNER-BOUND**
sinc	fixedk-2	3.409 [3.381, 3.440]	3.417 [3.393, 3.461]	+0.008 [−0.038, +0.065] **LEARNER-BOUND**
sinc	Poisson-2	3.239 [3.226, 3.264]	≥ 16	not scorable (lower bound)
comb	one spike	3.758 [3.738, 3.773]	≥ 16	not scorable (lower bound)
comb	fixedk-2	3.750 [3.736, 3.766]	≥ 8	not scorable (lower bound)
comb	Poisson-2	3.741 [3.729, 3.758]	≥ 16	not scorable (lower bound)

Two side readings travel with Table 1. First, the random sinc walls order as Poisson-2 3.239 < fixedk-2 3.409 ≈ one-spike 3.411: count *variability*, not spike density, lowers the smooth-kernel wall by ≈ 0.17, the sign the DC-mode derivation (T4.3) predicted at matched density (scored after the meta-review caught the omission; wording pending the close-out gate, no ledger row yet). Second,

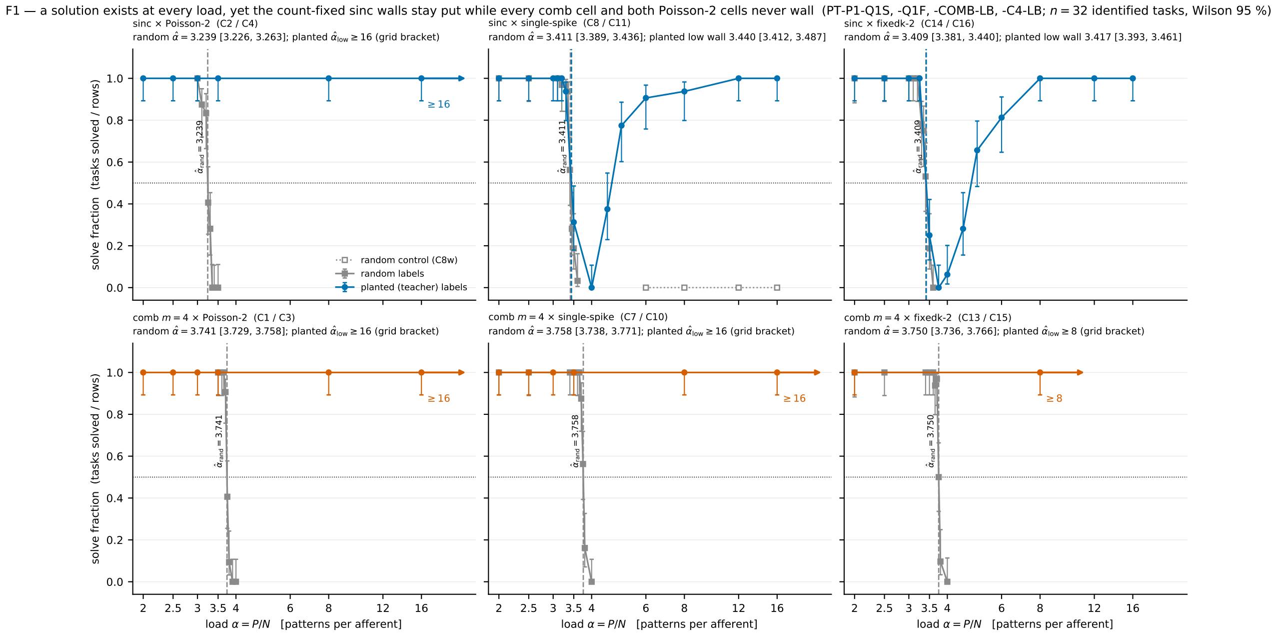

Figure 1: Solve fraction against load $\alpha = P/N$ for random labels (grey) and planted labels (colour) at $n = 32$ tasks per rung, Wilson 95 % intervals, for sinc (top) and comb (bottom) under Poisson-2, one-spike and fixedk-2 counts. Vertical lines mark the random half-crossings; arrows mark planted curves that never fall on their grid (lower bounds). Sinc: the planted curve tracks the random wall in both count-fixed ensembles and then re-solves above a hard band; under Poisson-2 it never falls. Comb: planted curves are flat at 1.0 to the grid top in every ensemble.

planting buys no speed: at every co-solved rung the median epochs-to-solve ratio between planted and random arms is within a factor 1.14 of one ($|\log r_{\text{med}}| \leq 0.13$; PR-10, no resolvable difference); the "50× planted speed-up" quoted in an early pilot was a load effect.

3.2 Re-solve and recovery (H-RECOVERY, Q4)

Above the wall the planted sinc curve returns. On the one-spike cell, with the seeds-8..31 random control bought as a gate correction, the window reads 29/32, 30/32, 32/32, 32/32 at $\alpha = 6, 8, 12, 16$ against 0/32 at every rung for random labels on the same pattern sets (paired 128/128 by spike hash), so the registered RE-SOLVE row fires from rung 6. Under fixedk-2 the shape repeats with a hard band at roughly $(3.5, 5)$: 8/32 at 3.5, 0/32 at 3.75, 2/32 at 4, 9/32 at 4.5, 21/32 at 5, 26/32 at 6 and 32/32 from 8. Figure 2 shows what the re-solved students are.

In the count-fixed ensembles the solutions found below and at the wall are unrelated to the planted teacher: median q between -0.02 and $+0.10$ from $\alpha = 2$ to 3.5, and the students' decision times coincide with the teacher's at chance ($A = 0.06$–0.09 against 0.067), for solved and unsolved tasks alike. Above the hard band every re-solved task has $q \geq 0.80$ (the largest low-load $|q|$ is 0.30): $0.93\,[0.89, 0.96]$ at $\alpha = 4.5$ rising to 0.999 at 16 on the one-spike cell, $0.87\,[0.85, 0.91]$ at 4.5 to 0.999 on fixedk-2. Under Poisson-2 counts the overlap rises gradually from $0.19\,[0.15, 0.22]$ at $\alpha = 2$ to 0.99 at 8, with neither wall nor plateau. The re-solve regime is teacher recovery, and the mechanism of the wall is that gradient descent finds *other* solutions there. The ordering of T5.1 holds: α_{onset} sits at 4.5 (one spike) and 4.0 (fixedk-2), above the random findable crossings of 3.41. The comb students land on $\pm w_T$ with $|q| \to 0.99$, the field-scale sign symmetry of a zero-mean kernel.

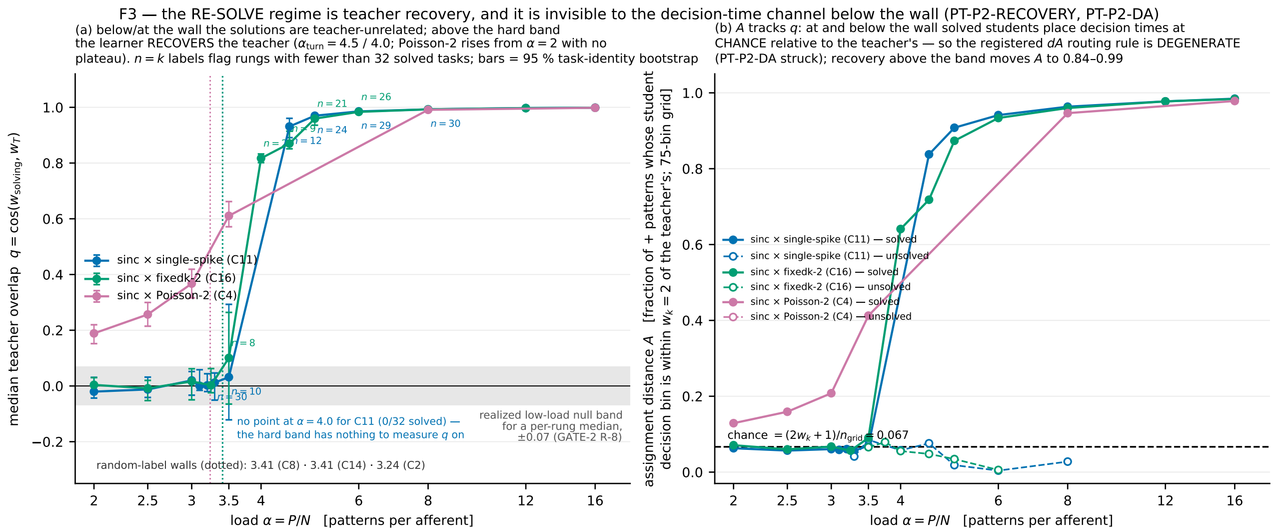

Figure 2: Left: median teacher overlap $q = \cos(w_{\text{solving}}, w_T)$ over solved tasks against α (log scale) for planted sinc under one-spike, fixedk-2 and Poisson-2 counts, bootstrap 95 % intervals, with the per-rung-median null band shaded and the random walls marked; n_{used} per point is annotated. Right: assignment distance A, the fraction of positive patterns whose student decision bin lies within two bins of the teacher's, for solved (filled) and unsolved (open) tasks; the dashed line is the chance level 0.067. Below and at the wall the solutions found are unrelated to the teacher; above the band they are the teacher.

3.3 The comb lift survives planting (Q3)

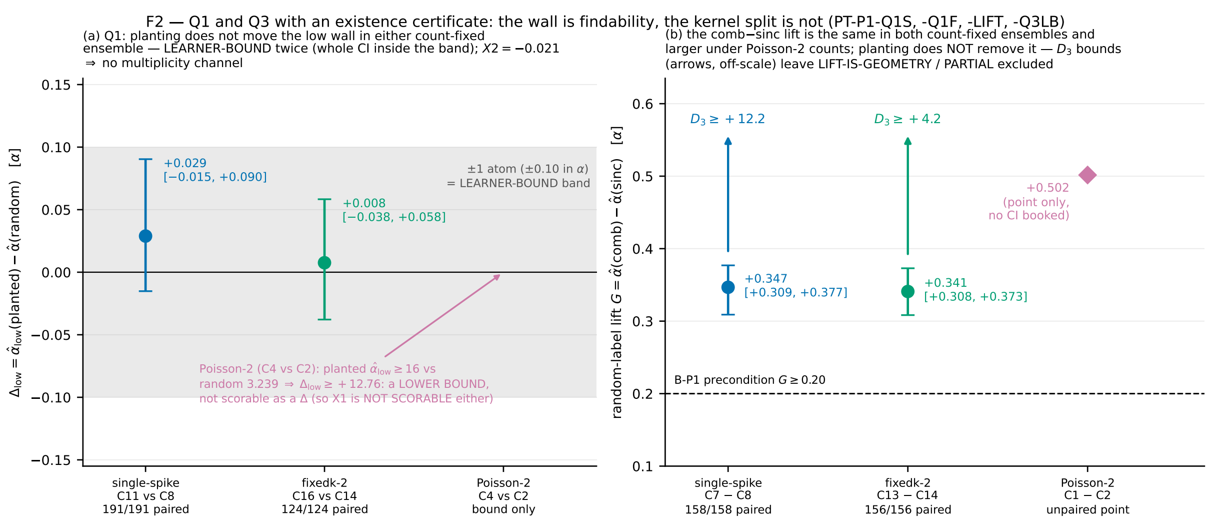

Figure 3: Left: the paired planted-minus-random wall difference Δ_{low} with 95 % intervals for the two count-fixed ensembles; the shaded band is ± 1 atom. Right: the paired comb-minus-sinc random lift G for the three ensembles and the Q3 lower bounds $D_{3,\text{lb}}$ that follow once the planted comb cells are read as grid brackets.

The random lift is $G = +0.347\,[+0.309, +0.377]$ (one spike, paired 158/158) and $+0.341\,[+0.308, +0.373]$ (fixedk-2, 156/156), indistinguishable between the count-fixed ensembles, and $+0.50$ under Poisson-2 (point, from the Phase 1 anchors). Since the planted comb curve never falls on its grid while planted sinc walls at 3.44 / 3.42, the registered bound form gives $D_{3,\text{lb}} = (T - \hat{\alpha}_{\text{low}}^{\text{sinc,planted}}) - G(\text{random})$ with

T the comb grid top: $+12.2$ (one spike, $T = 16$) and $+4.2$ (fixedk-2, $T = 8$). The kill row of the hypothesis that the lift is a learner-conditioning effect planting would remove ($D_3 < -0.35$) is not reached; LIFT-IS-GEOMETRY and PARTIAL are excluded in both count-fixed ensembles. These are lower bounds: the comb planted problem is at least this much easier, never "$D_3 = 12.2$". The between-ensemble discriminator registered alongside gives $X_2 = \Delta_{\text{low}}(\text{fixedk-2}) - \Delta_{\text{low}}(\text{one spike}) = -0.021$: no multiplicity channel at matched density.

3.4 The perturbation ladder (Q2, re-scoped)

The assignment-distance rule registered for Q2 was struck before any verdict: solved students place their decision times at chance relative to the teacher's, so "routing to the teacher" is not something the learner does and the rule's routing row could never fire. What remained was the ladder: initialise the student at $w_0 = w_T + r\|w_T\|\hat{g}$ with $\hat{g}$ an isotropic unit vector drawn per task, for $r \in \{0, 0.05, 0.1, 0.2, 0.4, 0.8, 1.6, 2.0, 4, 8\}$ and a random-init arm, at two loads per cell: the wall ($\alpha = 3.5$) and, at the user's choice, just above it (4.0 for the one-spike cell, 3.75 for fixedk-2), where random starts solve $0/32$ yet the teacher exists. The port was built additively, bit-identity tested and gated; every job pinned the GPU memory budget so that the learner's per-batch random streams are host-independent.

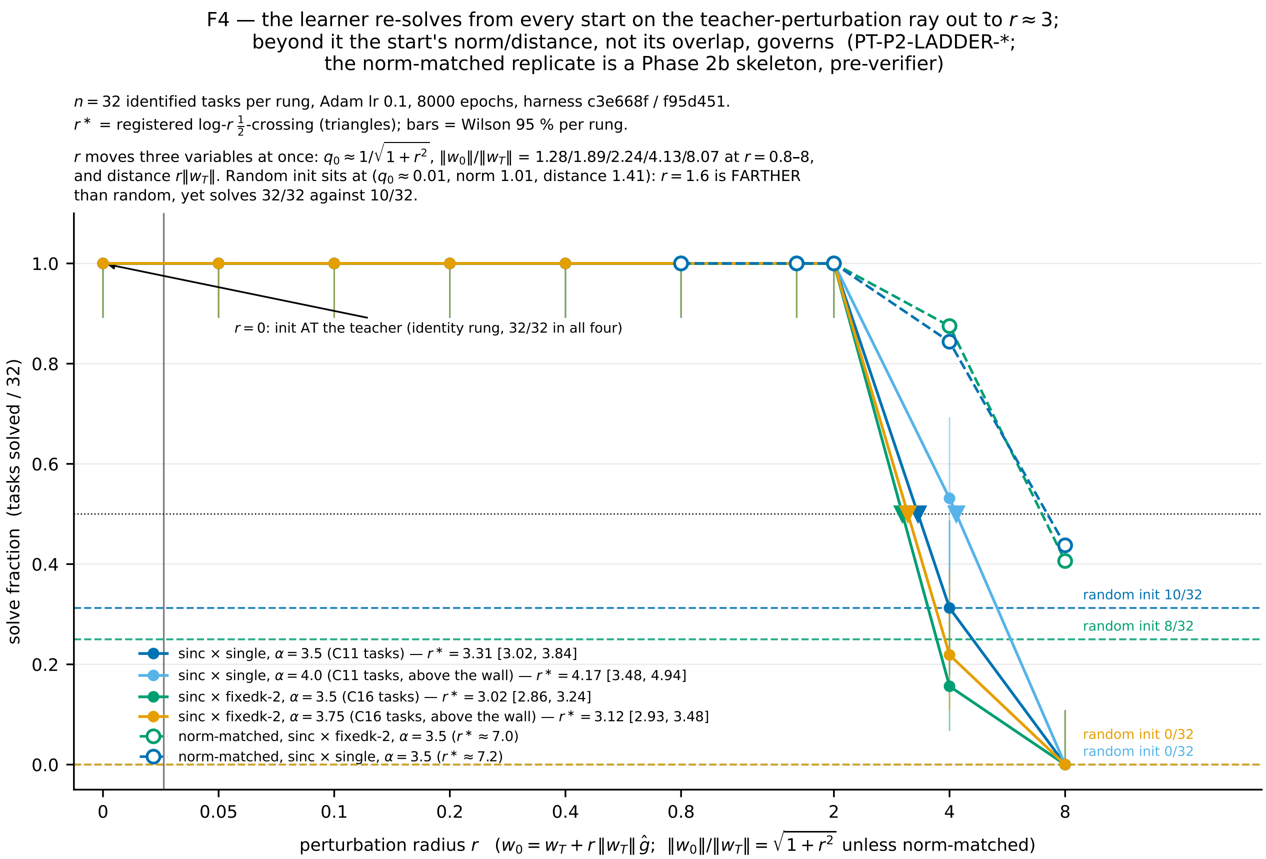

Figure 4: Solve fraction against perturbation radius r (log scale; $r = 0$ at the left) for the four (cell, load) ladders at $n = 32$, with the random-init rate as a dashed line per ladder and the registered log-r half-crossing r^* marked. Open markers: the norm-matched replicates on both cells at $\alpha = 3.5$ (Phase 2b, addendum-gated), where w_0 is rescaled to $\|w_T\|$ after the perturbation with the same per-task direction.

Table 2: Perturbation ladder: solved of 32 per radius; random-init arm; log-r half-crossing r^* with bootstrap interval (all rows gate-recomputed; the Phase 2b matched r^* has a bootstrap lower 2.5 % of 5.9 / 5.8 and an upper end not resolved by the grid). Above the wall the random arm for the one-spike cell is Phase 1's C11 rung 4.0 on the same pattern sets.

cell	α	$r \le 2.0$	$r = 4$	$r = 8$	random	r^*	bracket
sinc $\times$ one spike	3.5	32/32	10/32	0/32	10/32	3.3 $[3.0, 3.8]$	$(2, 4]$
sinc $\times$ one spike	4.0	32/32	17/32	0/32	0/32	4.2 $[3.5, 4.8]$	$(4, 8]$
sinc $\times$ fixedk-2	3.5	32/32	5/32	0/32	8/32	3.0 $[2.9, 3.2]$	$(2, 4]$
sinc $\times$ fixedk-2	3.75	32/32	7/32	0/32	0/32	3.1 $[2.9, 3.5]$	$(2, 4]$
fixedk-2, norm-matched (2b)	3.5	32/32	28/32	13/32	8/32	7.0	$(4, 8]$
one spike, norm-matched (2b)	3.5	32/32	27/32	14/32	10/32	7.2	$(4, 8]$

From every start on the perturbation ray up to $r = 2.0$ the learner re-solves 32/32, at the wall and above it, where random starts never solve (paired discordant 17/0 at $r = 4$ above the wall on the one-spike cell, exact $p = 1.5 \times 10^{-5}$). The half-crossing does not differ between the two loads within the ladder's factor-two resolution (ratios 1.26 $[0.97, 1.52]$ and 1.03 $[0.94, 1.17]$). Two readings are refused by the gate and are not made here: the word "basin", and "the learner returns to the teacher".

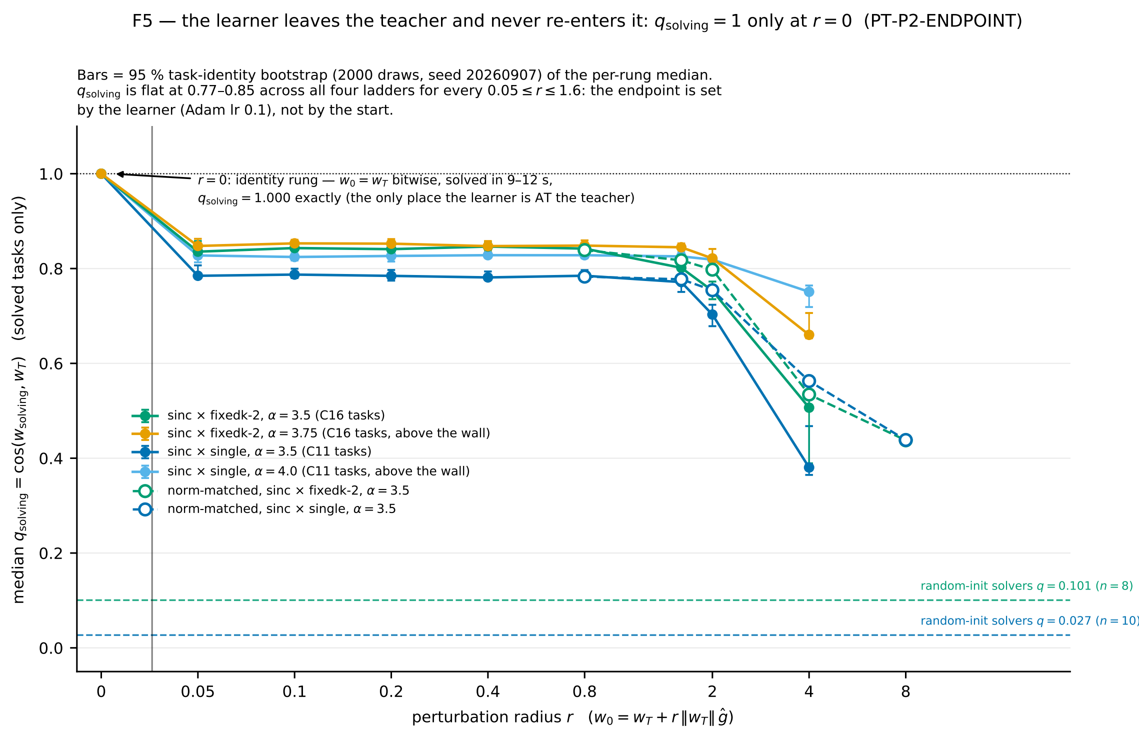

Figure 5: Median overlap of the solving weights with the teacher, q_{solving}, against r for the four ladders and the norm-matched replicate; the horizontal line is the random-init solvers' overlap; $r = 0$ is the identity point. The endpoint is the same overlap value $q \approx 0.8$ for every start with $r \le 1.6$.

The endpoints (Figure 5) show why. From every perturbed start with $0.05 \le r \le 1.6$ the learner *leaves* the teacher (it stays only at $r = 0$, where the solving weights equal w_T bitwise and the first check already reports solved) and re-solves after a median 1400–5400 epochs on a solution with $q \approx 0.77$–0.85 to the teacher and norm 3.5–8 $\|w_T\|$; 0.70–0.82 at $r = 2$; 0.38–0.75 at $r = 4$; random-init solutions sit at $q \approx 0.03$–0.10. The endpoint is set by the learner (Adam at learning rate 0.1 takes a first step of $\approx 0.1\,\|w_T\|$), not by the start. The TEACHER-ISOLATED flag is false in all four ladders, but it certifies only that the teacher's neighbourhood contains findable solutions, not that the teacher is an attractor.

The norm confound and its resolution. The perturbation $w_0 = w_T + r\|w_T\|\hat{g}$ moves overlap, distance and norm together: $\|w_0\| \approx \sqrt{1+r^2}\,\|w_T\|$, so the $r = 8$ start is eight times the teacher's norm while random init has norm $\approx \|w_T\|$. That $r = 8$ ($q_0 \approx 0.13$) solved 0/32, below the random rate with $q_0 \approx 0$, made the norm the suspect; the story that Adam could not shrink an $8\times$ field was refuted by the gate (unsolved $r = 8$ students end at 3–4 $\|w_T\|$). Phase 2b rescales w_0 to $\|w_T\|$ after the perturbation, with the same $\hat{g}$, under a registered three-row rule on the $r = 8$ count k_8 (NOT-NORM $k_8 \leq 3$, PARTIAL, NORM $k_8 \geq 8$; atom two tasks). On the fixedk-2 cell the norm-matched rows solve $32/32, 32/32, 32/32, 28/32, 13/32$ at $r = 0.8, 1.6, 2.0, 4, 8$: $k_8 = 13$ against 0 unmatched and 8 random, paired discordant 13/0, so the NORM row fires; the one-spike replicate repeats it ($32/32, 32/32, 32/32, 27/32, 14/32$, $k_8 = 14$ against 0 unmatched, paired discordant 14/0). The gated reading is precise about what this does and does not say. The un-matched collapse *below* the random-init rate at $r = 8$ was the initial norm, not the overlap on the teacher's sphere. But a matched start at $q_0 = 0.13$ buys nothing measurable over a random start: against the random rows on the same tasks the paired counts are 9/4 ($p = 0.27$) and 10/6 ($p = 0.45$), and 18–19 of 32 tasks still fail at $r = 8$. At $r = 4$ norm matching rescues 24 and 18 tasks (one reverse each), so the norm accounts for 23 of 27 and 17 of 22 un-matched failures there; the matched half-crossing moves to $r^* = 7.0$ and 7.2 (bracket $(4, 8]$, lower 2.5 % 5.9 / 5.8, upper end not resolved by the grid) against 3.0 / 3.3: the edge is partly a norm effect and moves out by about a factor two, it does not vanish. From a norm-one start the solutions still end at 5–9 $\|w_T\|$: the initial norm controls whether a low-overlap start solves, not where it ends. The two cells share their 32 teachers and rays bytes-for-bytes and differ in the spike patterns. The collapse at large r was the inflated starting norm; on the teacher's sphere the re-solving neighbourhood extends to an initial overlap of about 0.13, where it still beats random init. The ledger rows PT-P2B-NORM and PT-P2B-EDGE carry the endorsed wording.

3.5 Learner validity

Every number above carries three clauses measured in this campaign rather than assumed. **Learning rate (V4):** on random comb × Poisson-2 the wall moves by -0.069 at lr 0.05 and -0.017 at lr 0.2 (robust); on planted sinc × one-spike it moves $-0.189\,[-0.348, -0.105]$ at lr 0.2, and on the random arm of the same cell $-0.108\,[-0.148, +0.011]$, so the shift is not shown to be planting-specific and is treated as common mode within the rider's resolution. **Budget (V5):** doubling the anneal to 16 000 epochs moves the random sinc × Poisson-2 wall by $s = -0.017\,[-0.140, +0.120]$; solves land at the same fraction of the anneal, consistent with the schedule-timed picture of the August campaigns. **Gauge (V3):** the teacher's split-rule threshold is KS-equivalent to the random-mode gauge on 6/6 cells. **Anchors (V1, V2):** reproduced to ≤ 0.01.

4 Killed and struck

- **H-DC** (the comb lift is a Poisson DC-mode artefact and reverses in the one-spike ensemble): $G(\text{random, one spike}) = +0.28$ at $n = 16$, $+0.35$ at $n = 32$. The DC *sign* held on the smooth kernel (sinc's wall rises +0.17 from Poisson to count-fixed; comb immune).
- **H-DC-planted** for the comb (the planted Poisson ease is carried by the count-weighted feature $\sum_i w_{T,i} k_i$): that feature predicts comb-P planted labels at chance (AUC 0.502) while comb-P is the most findable planted cell. The zero-mean-teacher control was degenerate (it removes $1/N$ of the feature) and is booked as such.

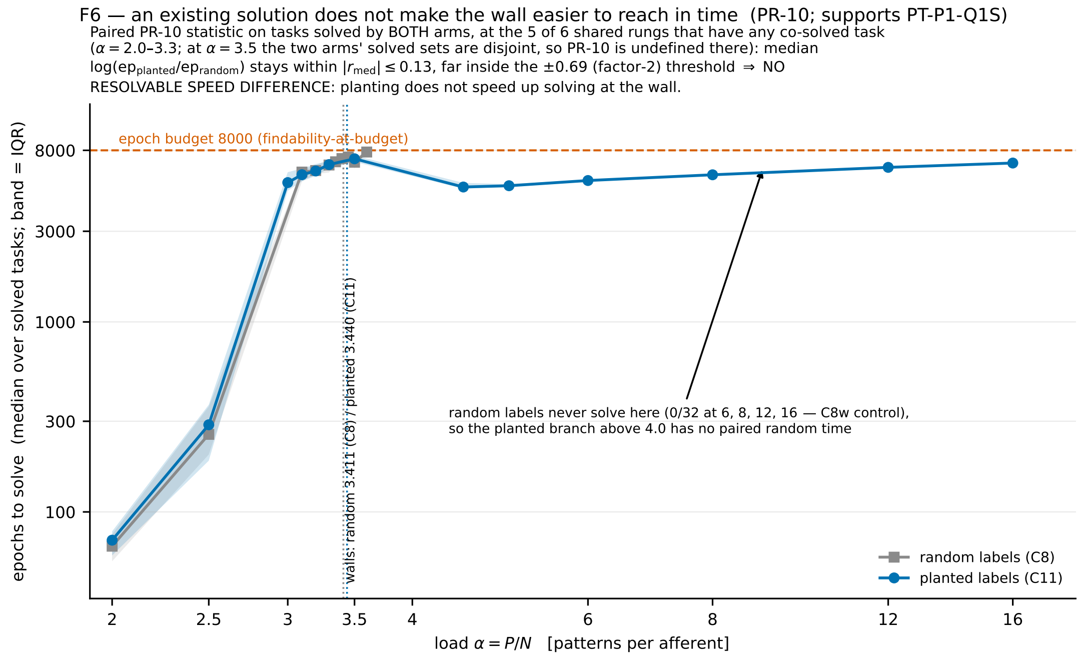

Figure 6: Median epochs to solve against α for random and planted labels on the one-spike sinc cells (C8, C11), interquartile bands. At co-solved rungs the two arms take the same time to within a factor 1.14: planting does not speed up solving at the wall.

- **H-COUNT** (a total-spike-count rate code): AUC 0.526, present and weak.
- **H-SCALE** premise (a per-pattern field-scale channel): the control predicted the labels equally well (0.938 vs 0.943); the kill clause had omitted the control's predicted value, a registration defect corrected by rule.
- **§1.1 assignment distance** (Q2's routing statistic): degenerate, solved students route to the teacher's timing at chance, so the routing row was unreachable.
- **"Learner phenomenon, not geometry"** as a fact, **"the lr sensitivity is the ensemble's"** as established, **"solutions sit on a shell"** as a geometric statement, **"basin"**, **"returns to the teacher"**: refused by the gates; weaker wordings stand.
- **Q4 order** (cross-kernel order of α_{onset}): not scorable, two kernels per ensemble put a full-order match at the $\frac{1}{4}$ floor. **dexp**: its random one-spike curve is non-monotone (a stuck sub-population) and the kernel left the grid.

5 Theoretical implications

1. **The wall behaves as an algorithmic wall.** At $N = 500$ the first place gradient descent stops finding solutions on half the problems does not move when a solution is planted, in two independent ensembles, to within 0.03. This closes the door the August campaigns left open: no capacity claim of this program can rest on a findability wall, and the size-biasing bonus of T1 acts on existence, which the learner does not see.
2. **A hard phase, then recovery.** The planted curve has the easy/hard/easy shape of planted constraint-satisfaction problems: below the wall the learner finds one of the exponentially many

uncorrelated solutions; in the band it finds nothing in budget; above it the uncorrelated solutions have gone and the only thing left to find is the teacher, which it then recovers to $q = 0.999$. The onset sits above the random findable crossing, as T5.1 requires, and the recovery load is a learner-side estimate of where the random-label solution space empties: between $\alpha \approx 4.0$ and 4.5 for sinc at $K_{\text{eff}} = 32$, roughly one unit above the findable wall. Under Poisson counts the uncorrelated solutions never dominate: q rises from $\alpha = 2$, the count fluctuation having made the teacher's labels partly a low-dimensional code.

3. **The comb lift is not learner conditioning.** Planting removes the sinc wall's mystery by relocating it to the learner, but the comb has no low wall at all, planted or random beyond 3.75; the split between kernels is a property of the problem the kernel poses, not of what planting removes from the learner's task. Its Poisson excess (+0.50 against +0.35) is the DC mode acting on sinc.

4. **The teacher is not an attractor; its neighbourhood is.** Gradient descent leaves the planted solution at the first step and settles on a nearby solution of larger norm and overlap ≈ 0.8; it does so from every start whose initial overlap is above roughly 0.13 once norm is controlled, and never from random starts above the wall. The geometry a learner sees is therefore a connected re-solving region around the teacher's direction, with random initialisation lying outside it, and once norm is controlled the edge sits near an initial overlap of 0.13–0.25, where a start is no better than random: the picture of a hard phase in which what a start buys is its overlap with the teacher, with the starting norm deciding only whether a low-overlap start solves at all.

6 Key takeaways

- Findability walls at $N = 500$ behave as learner walls: planted and random low walls agree to < 1 atom in both count-fixed ensembles.
- The planted problem re-solves above a hard band, and the re-solved student is the teacher ($q \geq 0.80$ on every re-solved task); the solutions found at the wall are not.
- The comb–sinc split survives planting ($D_3 \geq +12.2$ and $\geq +4.2$); the random lift is +0.35 / +0.34 / +0.50 across ensembles; count variability, not density, moves the smooth-kernel wall.
- Around the teacher the learner re-solves from any start within $r \leq 2$, at and above the wall; with the starting norm matched the edge moves out about twofold ($r^* \approx 7$) and a start at overlap 0.13 is no better than random.
- Four mechanistic hypotheses and one rider statistic died on registered kills (two instruments degenerate); the campaign's own process caught a false sampling-frame sentence, a mislabelled pooled statistic, a per-batch learner RNG, and an unscored registered discriminator.

7 Next steps

1. **Locate the emptying load.** The recovery onset brackets $(3.5, 4.5]$ (one spike) and $(3.75, 4.0]$ (fixedk-2). A denser planted grid in those brackets, with the q readout, gives a learner-side estimate of the random-label existence capacity at $N = 500$ that can be set against the replica surrogate with the count law included ($g = 0$ and $g = 1$).

2. **A matched-norm ladder above the wall** (both cells were matched at the wall only), to state the overlap edge in both ensembles and both loads.

3. **Why Poisson recovers early.** The DC-mode derivation predicts a low-dimensional label code under count fluctuation; the AUC channels tested were weak. A direct test is the teacher's projection onto the count mode versus q at low load.

4. **The comb's missing wall.** Planted comb never walls to $\alpha = 16$; buying $\alpha = 24$ (the costed gap) and a planted comb ladder would show whether the comb's students also recover the teacher or find other solutions indefinitely.

5. **Harness.** Record `s_budget` and the GPU in `results.json`; seed the learner's streams per task, not per batch; checkpoint per seed batch so a killed rung resumes; stream finished rungs to the object store. The last two would have saved $\approx$ \$12 of the \$136 spent.

8 Methods and process appendix

8.1 Statistics

Half-crossing: first downward linear-interpolated crossing of the solve fraction through $\frac{1}{2}$ scanning upward in α; a curve $\geq \frac{1}{2}$ at every rung is a grid bracket and is never written as a crossing. Intervals: bootstrap over task identities (2000 draws, seed 20260904), jointly resampled for paired statistics; `boot_finite` reported. Two interval conventions coexist: the figures plot the committed scorer's intervals, the text and ledger quote the gate's independent recomputation; edges differ by ≤ 0.007 in α (0.11 on one r^* edge). Paired contrasts require identical spike hashes per (seed, α) across arms. Exclusions (teacher validity acc $= 1$ and no $V_{\max}$ ties; solved $\Rightarrow$ final error 0 and CSV/npz agreement; random label balance in $[0.47, 0.53]$) are identity-logged and applied to both arms. Every decision rule was registered as a JSON block with thresholds, a resolution atom, null and effect rows, a degenerate world mapping to a non-verdict, and preconditions with consequences, and checked mechanically before spend (`tools/claim_check.py plan`); every claim file was checked after (`claims`); the ledger validates with 144 checks and 0 failures.

8.2 Verification and gates

Seven gates (GATE-0, GATE-1, GATE-1b, the riders' GATE-1, two GATE-2s, GATE-2b) by an independent verifier agent recomputed every booked number from the raw rows (all to $\leq 10^{-6}$), checked pairing hashes, exclusion blindness, the bound-form identity to 10^{-15}, and the port diffs (bit-identity of every pre-existing output with the new keys absent). Weakenings the gates forced are quoted in the ledger rows and repeated in the text above. A meta-review at the Phase 1/2 boundary ranked the recurring shapes: rules that pass the registration lint yet cannot attain their effect row on landed data (fourth campaign running); a registered discriminator bought and unscored (X_2); job wall-clock never modelled; kill clauses without the control's predicted value; pairing asserted from arguments instead of hashes.

8.3 Catches worth recording

The task-identity rule "(seed, α) regardless of stream length" was verified for the count-fixed ensembles (identical spike hashes 48/48, 64/64) and refuted for Poisson by a direct test (8/8 hashes differ across stream lengths, the `max_spikes` path): the Phase 1 Poisson anchors are independent replications, not replays. The learner's random streams (init, restarts, shuffle) are seeded per seed *batch*, and the batch size follows the GPU memory budget, so a result is bit-regenerable only at the same budget; pinning the budget fixed a CUDA out-of-memory in the comb kernel's trace precompute (the Phase 0 large-α comb cells had run on 102 GB cards) at the price of 2–3$\times$ longer jobs and three zero-row timeouts before one-rung-per-job became the rule. The scorer's "pooled" K_{eff} was a min-per-rung statistic; the scheduler snapshots the harness HEAD at launch and its clock runs a day behind.

8.4 Spend and operations

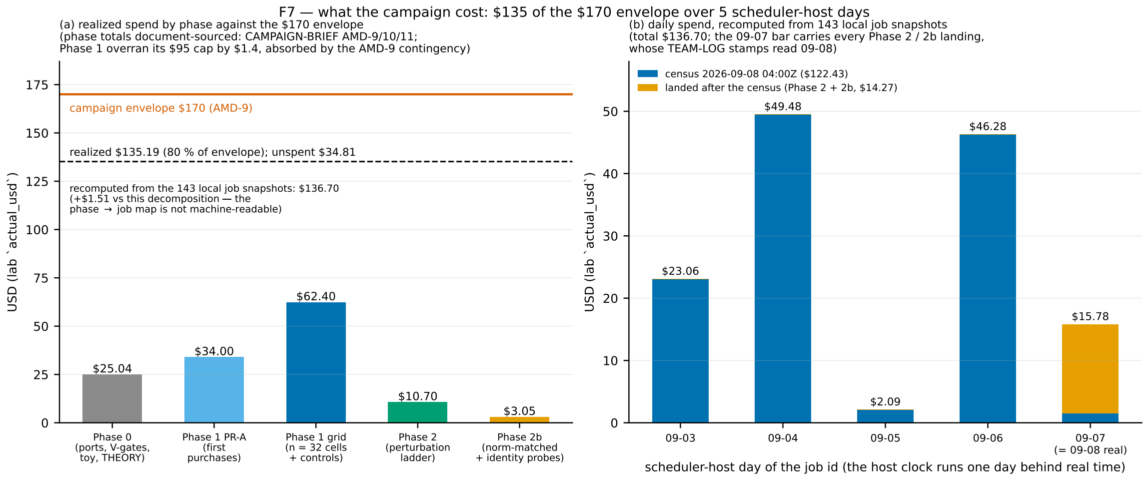

Figure 7: Realized spend by phase against the $170 envelope (left) and by scheduler-host day (right). The Phase 1 grid exceeded its cap by ≈ $1.4 after three zero-row timeouts and one bought control; Phase 2 and 2b came in under.

Realized spend: Phase 0 $25.04; Phase 1 ≈ $96.4 against a $95 cap (PR-A $34.0, grid $62.4 including a $8.9 control bought as a gate correction and $10.4 of zero-row timeouts); Phase 2 $10.7; Phase 2b $3.6 (two landed jobs $1.14; one anomalous scheduler run lost $2.4); total ≈ $136 of $170 across ≈ 340 job ids. Dead hosts at provisioning cost nothing but dominated the worst waves (80–90 % of launches; 95 of 147 hand submissions in Phase 1 never ran); one host without a working GPU recurred; two healthy jobs were cancelled by the orchestrator after misreading the scheduler host's clock (≈ $2). The queue was run for two days by a Sonnet-class operations agent under a fixed rulebook, with landings scored mechanically and recomputed at the gates.

8.5 Reproducibility and availability

Harness: `projects/tempotron-capacity`, branch `campaign/2026-09-planted-teacher`, tags `campaign-2026-09-` (`2632a91`), `-phase2` (`c3e668f`), `-phase2b` (`f95d451`; the one-spike norm-matched job ran at `8aba5cc`, a tests-only commit above it with the harness bytes unchanged). Every job's command line, seed list, job id and cost is in the workstream team logs; every scored claim is a JSON file under `data/`; raw `results.csv` rows and saved weights are in the run directories and mirrored in the object store. Figures are generated by `report/make_figures.py` from those files; `FIGURES.md` lists each figure's provenance. Statements: no human or animal data; no external funding beyond the user's compute budget; no competing interests; the text, analysis code and figures were produced by Claude-family agents under the user's direction, with every numerical claim recomputed by an independent verifier agent from the raw rows.

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